

SE Lab 04 Climber Game

(TASKS AND CHANGES MADE IN CODE)

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Task 1: Fix the height-score bug

Bug:

```
current_height = max(0, (HEIGHT - 40 - self.player.rect.y) // 10)
self.height = current_height
```

Fix:

```
current_height = max(0, (HEIGHT - 40 - self.player.rect.y) // 10)
self.height = max(self.height, current_height)
```

Explanation:

The old code replaces the stored height with the player's current height every frame. Therefore, when the player moves downward, the displayed Height also decreases. The updated code keeps the larger value between the previous best height and the current height, so the Height records the maximum height reached during the run.

Task 2: Implement progressive platform colours

Bug:

```
def platform_color(index, total):
    pass
```

Fix:

```
def platform_color(index, total):
    if index == 0:
        return None

    progress = index / max(1, total)
    r = int(100 + 100 * progress)
    g = int(180 - 60 * progress)
    b = int(100 + 100 * progress)
    return (r, g, b)
```

Explanation:

The old function did not return a colour, so platforms used the default green colour. The updated function uses the platform's index relative to the total number of platforms to gradually change the RGB values. The ground platform keeps the default green colour while higher platforms receive different shades.

Task 3: Add moving platforms

Bug:

```
def moving_platform_speed(index, total):  
    pass
```

Fix:

```
def moving_platform_speed(index, total):  
    if index % 3 == 0:  
        return 1.5  
    return 0
```

Explanation:

The old function returned nothing, so all platforms remained stationary. The updated function gives every third platform a small horizontal speed of 1.5 pixels per frame, while the remaining platforms return 0 and stay static. The existing Platform.update() method already handles the horizontal oscillation and direction reversal.

Task 4: Add a sparkle effect after collecting a coin

Bug:

```
def on_coin_collected(coin, score):  
    pass
```

Fix:

```
sparkles = []  
  
def on_coin_collected(coin, score):  
    for _ in range(8):  
        speed = random.uniform(1.5, 3.0)  
        sparkles.append({  
            "pos": pygame.Vector2(coin.pos),  
            "vel": pygame.Vector2(  
                random.uniform(-speed, speed),  
                random.uniform(-speed, speed)  
            ),  
            "life": 20  
        })
```

Supporting update and draw logic:

```
for sparkle in sparkles[:]:  
    sparkle["pos"] += sparkle["vel"]  
    sparkle["life"] -= 1  
  
    if sparkle["life"] <= 0:  
        sparkles.remove(sparkle)  
  
for sparkle in sparkles:  
    x = int(sparkle["pos"].x)  
    y = int(sparkle["pos"].y - self.cam_y)  
    radius = max(1, sparkle["life"] // 5)
```

```
pygame.draw.circle(  
    screen, (255, 240, 120), (x, y), radius  
)
```

Explanation:

The old function did nothing after a coin was collected. The updated implementation creates several short-lived sparkle particles at the coin's position. Their position and lifetime are updated every frame and they are drawn briefly before being removed. The existing 50-point coin scoring and collision logic remain unchanged.